

# HEALTHCARE MANAGEMENT: PATIENT READMISSION PREDICTION

## Project Overview

This capstone project analyzes 101,766 diabetic patient encounters across 130 U.S. hospitals (1999–2008) to predict the likelihood of unplanned 30-day readmissions. The project leverages a multi-tool approach with Excel, SQL, Python, and Tableau to show actionable insights for hospital administrators, clinical care teams, and healthcare policy makers. The dataset, sourced from the UCI Machine Learning Repository, contains 50 features per encounter, offering a rich foundation for both exploratory and predictive analysis.

## Methodology

Data quality steps taken before analysis, including:

- Excluding 3 records with invalid gender values
- Labeling 2,273 missing race values as 'Unknown/Not Reported'
- Dropping the weight field due to over 90% missing data
- Filtering '?' placeholder values from diagnosis fields in SQL queries

## Key Findings

- 30-Day Readmission Rate: 11% of patients (11,357) were readmitted within 30 days, representing the highest-risk clinical group.
- Age as a Risk Factor: Patients aged 60–80 account for the highest volume of encounters and the highest 30-day readmission rates across all age groups.
- Emergency Admissions: Over 53,000 emergency admissions were recorded, carrying the longest average hospital stays and elevated readmission risk.
- Top Diagnoses: ICD-9 codes 428 (Heart Failure) and 414 (Coronary Artery Disease) are the most common primary diagnoses, both conditions strongly associated with readmission.
- Medication: Patients averaged 16 medications per encounter, with higher counts correlating with longer stays and greater readmission risk.
- Payer Disparities: Readmission rates vary by insurance type, with Self Pay patients showing notably elevated 30-day readmission rates.

## Recommendations

- Implement a discharge risk scoring tool to flag high-risk patients; those with multiple medications, prior inpatient visits, or emergency admissions, for intensive post-discharge follow-up.
- Develop age-targeted care transition protocols specifically for patients aged 60–80.
- Investigate payer-specific readmission disparities to determine whether gaps in post-discharge coverage are contributing to elevated rates among Self Pay patients.
- Expand data collection to include social determinants of health (housing, income, transportation) to improve future predictive model accuracy.

## Conclusion

This project demonstrates how structured data analysis across multiple tools can translate a large clinical dataset into clear, actionable insights. By identifying the patient profiles most likely to experience unplanned readmissions older patients with complex diagnoses, high medication loads, and emergency admission types healthcare systems can shift from reactive to proactive care delivery. Implementing the recommended interventions has the potential to reduce readmission rates, lower costs, and meaningfully improve patient outcomes.